# Mehdi Zafari

**Ph.D. Candidate, UC Irvine**

Signal processing and AI/ML for next-generation (5G/6G) wireless systems

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Updated October 2026

## Education



### Ph.D. in Electrical Engineering and Computer Science

**June 2024 – December 2028, expected**

University of California, Irvine, CA, USA

GPA: 4/4

Advisor: Prof. A. Lee Swindlehurst



### M.S. in Electrical and Computer Engineering

**August 2021 – May 2024**

Rice University, Houston, TX, USA

GPA: 3.92/4

Thesis: Distributed Multi-User Beamforming in the Presence of CFO: An Analytical and Experimental Study



### B.S. in Electrical and Computer Engineering

**September 2015 – September 2019**

University of Tehran, Tehran, Iran

GPA: 3.7/4

Thesis: Design and Hardware Implementation of a MIMO-OFDM System Using High-Level Synthesis Techniques

## Research interests

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|---|---|
| **Wireless communications** | Cell-Free MIMO, Integrated Sensing and Communication, Resource Allocation, Beamforming |
| **ML and optimization** | AI/ML for Wireless, Distributed Optimization, Generative AI for Sensing and Wildfire Prediction |

## Experience

**Jul 2024 – present**

### Graduate Student Researcher

EECS, UC Irvine

- Conduct research on reinforcement learning and graph neural network (GNN)-based models for cooperative ISAC networks
- Design distributed algorithms based on ADMM/consensus for beamforming and resource allocation in cell-free ISAC systems
- Develop large-scale network simulations and implement optimization/learning pipelines using Python/PyTorch, CVXPY, Gurobi
- Supervisor: Prof. A. Lee Swindlehurst

**Jun – Sep 2026**

### Wireless AI Engineering Intern

Tiami Networks, Sacramento, CA

- Investigated ISAC channel models and constructed high-fidelity RF digital twins using NVIDIA Sionna and ray-tracing engines
- Analyzed spatial propagation characteristics to optimize sensing algorithms and facilitate integration with Open RAN architectures

Aug 2021 – Apr 2024 **Graduate Research Assistant** ECE, Rice University

- Conducted research on distributed multi-user MIMO beamforming for next-generation wireless networks
- Performed over-the-air experiments using the RENEW massive MIMO testbed to evaluate distributed multi-user beamforming
- Published four research papers and released over 200 datasets collected via the RENEW platform
- Datasets: Distributed Multi-User MIMO Datasets on IEEE DataPort
- Supervisor: Dr. Rahman Doost-Mohammady

Jan – Apr 2024 **Research Fellow** Rice D2K Lab

- Mentored five undergrad students on a Rice EMS-sponsored project to analyze operational data across multiple domains
- Developed time-series prediction models to anticipate future requirements in each area

Dec 2018 – Dec 2020 **Undergraduate Research Assistant** Array Processing Lab, University of Tehran

- Conducted research on the hardware implementation of MIMO OFDM systems using the Xilinx ZedBoard platform
- Designed full-stack OFDM system, from theoretical modeling in MATLAB/Python to hardware implementation via Vivado-HLS
- Developed OFDM and SC-FDE systems on Xilinx FPGAs using High-Level Synthesis (HLS)
- Supervisor: Dr. Aliazam Abbasfar

May – Nov 2020 **Part-Time R&D Engineer** Transportation Innovation Center, Tehran

- Designed and implemented hybrid radio links (FM analog and OFDM digital) for radio-controlled camera drones
- Researched telemetry system implementation strategies for vehicular communication in drone networks

May – Nov 2020 **Part-Time R&D Engineer** Sadr Electronic Dena, Tehran

- Developed a 2.4GHz ISM band IoT system to enable remote control functionality for elevator central control boards
- Engineered embedded C/C++ firmware for AVR and ARM microcontrollers with serial data transmission protocols

Jul – Oct 2018 **Full-Time Intern** Pionaria, Tehran

- Simulated parabolic and patch antennas in HFSS and CST, improving VSWR and gain
- Implemented point-to-point and point-to-multipoint wireless links using wireless network standards and software-defined radio (SDR) concepts

## Publications

### Journal and magazine papers

**CORDIS: A Scalable Coordinated Resource Allocation Framework for Distributed Cell-Free ISAC**

**M. Zafari**, B. Ottersten, and A. L. Swindlehurst

IEEE TWC Submitted to IEEE Transactions on Wireless Communications

Under review

**Robust Wildfire Forecasting under Partial Observability: From Reconstruction to Prediction**

C. Yang, **M. Zafari**, Z. Duan, and A. L. Swindlehurst

IEEE TGRS Submitted to IEEE Transactions on Geoscience and Remote Sensing

Under review

**Multi-Domain Optimization Framework for ISAC: From Electromagnetic Shaping to Network Cooperation**

R. Liu, M. Li, **M. Zafari**, B. Ottersten, and A. L. Swindlehurst

IEEE WCM IEEE Wireless Communications (magazine)

**An Analytical and Experimental Study of Distributed Uplink Beamforming in the Presence of Carrier Frequency Offsets**

**M. Zafari**, D. Pandey, and R. Doost-Mohammady

IEEE TVT IEEE Transactions on Vehicular Technology

## Conference papers

**TiamiTwin: A Digital Twin for Bistatic ISAC Drone Sensing, Validated Against Measurements**

**M. Zafari**, S. Enayati, A. L. Swindlehurst, and A. Mukherjee

MILCOM 2026 Workshop IEEE MILCOM 2026 Workshop on Integrated Sensing and Communication for Critical Infrastructure Protection (ISAC4CIP)

Accepted

**ASSENT: Learning-Based Association Optimization for Distributed Cell-Free ISAC**

**M. Zafari** and A. L. Swindlehurst

ICC 2026 IEEE International Conference on Communications (ICC)

**Coordinated Decentralized Resource Optimization for Cell-Free ISAC Systems**

**M. Zafari**, R. Liu, and A. L. Swindlehurst

Asilomar 2025 59th Asilomar Conference on Signals, Systems, and Computers

Best Paper Award Finalist

**ADMM for Downlink Beamforming in Cell-Free Massive MIMO Systems**

**M. Zafari**, D. Pandey, R. Doost-Mohammady, and C. A. Uribe

Asilomar 2024 58th Asilomar Conference on Signals, Systems, and Computers

**ML-Based Feedback-Free Adaptive MCS Selection for Massive Multi-User MIMO**

Q. An, **M. Zafari**, C. Dick, S. Segarra, A. Sabharwal, and R. Doost-Mohammady

Asilomar 2023 57th Asilomar Conference on Signals, Systems, and Computers

**Robustness of Distributed Multi-User Beamforming: An Experimental Evaluation**

R. Doost-Mohammady, **M. Zafari**, and A. Sabharwal

SAM 2022 IEEE 12th Sensor Array and Multichannel Signal Processing Workshop (SAM)

## Datasets and software

**Deep-Gen-Project: Spatiotemporal Generative Models for Wildfire Spread Prediction**

**M. Zafari**, E. Finkelstein, and C. Yang

GitHub GitHub repository

**ASSENT-CellFree-ISAC: Simulation Framework**

**M. Zafari**

GitHub GitHub repository

**Distributed Cell-Free ISAC Simulation Framework**

**M. Zafari**

GitHub GitHub repository

**Distributed Multi-User MIMO Datasets**

**M. Zafari**

IEEE DataPort IEEE DataPort

## Talks

Mar 2026 Competition talk

**When Wireless Signals Become Our Eyes**

UCI Grad Slam finals, Irvine, CA

3rd place.

Oct 2025 Lecture

**Coordinated Decentralized Resource Optimization for Cell-Free ISAC Systems**

59th Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA

Best Paper Award finalist; also presented a poster in the Student Paper Contest.

Oct 2024 Lecture

**ADMM for Downlink Beamforming in Cell-Free Massive MIMO Systems**

58th Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA

Oct 2023 Poster

**ML-Based Feedback-Free Adaptive MCS Selection for Massive Multi-User MIMO**

57th Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, CA

Sep 2023 Poster

**Distributed optimization for beamforming in massive MIMO systems**

Texas Colloquium on Distributed Learning (TL;DR 2023), Rice University, Houston, TX

Jan 2023 Project presentation

**DNN-based feedback-free adaptive modulation for massive MIMO**

POWDER-RENEW Mobile and Wireless Week, University of Utah, Salt Lake City, UT

1st prize for the project.

## Honors and awards

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| 2026 | **Wireless History Foundation Scholarship Award** | Wireless History Foundation |
| 2026 | **Pia Gratton Endowed Graduate Student Award** | UC Irvine |
| 2026 | **1st Place, UCI EECS Annual Technology Showcase** | UC Irvine |
| 2026 | **3rd Place, UCI Grad Slam Competition** | UC Irvine |

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| 2025 | **Best Paper Award Finalist, 59th Asilomar Conference** | IEEE Asilomar Conference |
| 2024 | **Graduate Dean's Recruitment Fellowship Award** | UC Irvine |
| 2022, 2023 | **SIAWE Scholarship Award for Distinguished International Students**<br>Society of Iranian-American Women for Education (SIAWE). | Rice University |
| 2023 | **1st Prize, POWDER-RENEW Mobile and Wireless Week** | University of Utah |
| 2021 | **Rice University Graduate Fellowship** | Rice University |
| 2019 | **Exceptional Talent Award**<br>Top 5% GPA. | University of Tehran |
| 2016 | **National Elites Foundation Fellowship** | Iranian National Elites Foundation |
| 2015 | **Ranked 47th, Iranian National University Entrance Exam (Konkoor)** | |
| 2013 | **Ranked 3rd, Iranian National Mathematics Olympiad** | |

## Service

2024–2026 **Peer reviewer**

- IEEE Transactions on Wireless Communications (TWC)
- IEEE Transactions on Vehicular Technology (TVT)
- IEEE Transactions on Geoscience and Remote Sensing (TGRS)
- IEEE Transactions on Aerospace and Electronic Systems (TAES)
- IEEE Wireless Communications Letters (WCL)
- IEEE Signal Processing Letters (SPL)
- IEEE Communications Magazine

## Teaching

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|---|---|---|
| Fall 2026 | **Introduction to Control Systems + Lab** EECS 160A/LA | UC Irvine |
| Spring 2025 | **Introduction to Digital Logic Laboratory** | UC Irvine |
| Spring 2024 | **Applied Machine Learning & Data Science** | Rice University |
| Fall 2023 | **Modern Communication: Theory and Practice** | Rice University |
| Fall 2022 | **Modern Communication: Theory and Practice** | Rice University |
| Spring 2020 | **Wireless Multimedia Communications** | University of Tehran |
| Fall 2019 | **Signals and Systems Analysis** | University of Tehran |
| Spring 2019 | **Digital Communications** | University of Tehran |

Teaching assistant for every course listed.

## Skills

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| **Key skills** | Wireless Communications, Signal Processing, Optimization, Applied Machine Learning |
| **Programming** | C++, Python, TensorFlow, PyTorch, CVXPY, Gurobi, MATLAB, Verilog |
| **Software-defined radios** | USRP B210/X410, NVIDIA Aerial, GNU Radio, Iris SDR, Agora |

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| **Simulation tools** | NVIDIA Sionna, Ray-Tracing Engines, ns-3, Wireshark, Ansys HFSS, Simulink |
| **Embedded and FPGA** | ARM STM32, DSP, Atmel AVR, Xilinx ZedBoard, Vivado HLS |

## Relevant coursework

| | | |
|---|---|---|
| Fall 2025 | **Deep Generative Models**<br>Grade: A | UC Irvine |
| Fall 2024 | **Machine Learning for Signal Processing**<br>Grade: A+ | UC Irvine |
| Spring 2024 | **Optimization: Algorithms, Complexity, & Approximations**<br>Grade: A | Rice University |
| Spring 2023 | **Distributed Methods for Optimization and Machine Learning**<br>Grade: A+ | Rice University |
| Fall 2022 | **Machine Learning (Introduction to Deep Learning)**<br>Grade: A | Rice University |
| Spring 2022 | **Advanced Wireless Networks**<br>Grade: A | Rice University |
| Fall 2021 | **Modern Communication: Theory and Practice**<br>Grade: A | Rice University |
| Fall 2017 | **Advanced Object-Oriented Programming**<br>Grade: A | University of Tehran |

## Selected conferences and workshops

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|---|---|---|
| Oct 2025, 2024, 2023 | **Asilomar Conference on Signals, Systems, and Computers** | Pacific Grove, CA |
| Oct 2025 | **IEEE Military Communications Conference (MILCOM)** | Los Angeles, CA |
| Sep 2023 | **Texas Colloquium on Distributed Learning** | Houston, TX |